

INVESTOR RESILIENCE

National Hackathon — Problem Statement

Organised by: Club of ProgrammerS (COPS), IIT (BHU) Varanasi **In collaboration with:** Securities and Exchange Board of India (SEBI) & National Securities Depository Limited (NSDL) **Format:** 7-Day Build Sprint | 5 focus tracks + 1 open track

1. The Challenge

India's retail investor base has grown at an unprecedented pace: total Demat accounts have crossed 16+ crore, with more than 70% of incremental retail accounts opening from non-metro, Tier-2, and Tier-3 cities. Access to markets has outrun access to financial confidence.

For most first-time and retail investors, financial decisions are shaped by misinformation, fraudulent schemes, social-media "tips," emotional decision-making, complex terminology, or simply the absence of trustworthy guidance. SEBI's own studies show that 9 out of 10 individual traders in equity F&O incur net losses — a signal that market access alone is not building resilience.

The challenge, therefore, is not to help people invest — it is to help them become resilient investors: people who can identify risk, avoid fraud, understand financial products, build healthy habits, and decide independently and safely.

The objective is not to help users earn more money. It is to help users lose less, understand more, behave better, and decide more safely.

2. Mission

Build a technology-driven product that strengthens the financial resilience of Indian investors, with a particular focus on users from Tier-2 and Tier-3 cities and emerging digital-finance users. The product should address a real problem an investor may face before, during, or after a financial decision.

3. Core Design Principle

A strong solution should answer at least one of the following. The strongest solutions connect several of these into a single, coherent product.

- Help an investor detect a financial fraud or scam before becoming a victim
- Help an investor recognise misleading or manipulative financial information
- Explain a financial product or regulatory process in simple, plain language
- Build better financial habits and more deliberate decision-making
- Reduce panic, greed, herd behaviour, or impulsive financial decisions
- Clarify an investor's rights, responsibilities, and protection mechanisms

- Help an investor recover intelligently and safely after something goes wrong
 - Build financial confidence without creating dependency on "tips" or influencers
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4. Mandatory Guardrails — What Not to Build

This is a public-good hackathon, not a fintech pitch competition. Submissions that violate the following will be disqualified regardless of technical quality:

- **No commercial or speculative outcome** — no stock tips, buy/sell/hold signals, price predictions, trading algorithms, or promotion of a specific instrument or broker.
 - **No monetisation funnels** — no broking commissions, margin-financing nudges, or paid subscription upsells directed at the user.
 - **Privacy by design** — no unauthorised harvesting of SMS, OTPs, or personally identifiable financial records.
 - **Public-good ethos** — the product should read as investor-protection infrastructure, not a growth product.
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5. Challenge Tracks

Teams may pick one track, combine tracks, or enter the Open Track. Depth on one real user journey is valued over breadth across many features.

Track A — Digital Fraud & Scam Resilience

Focus: Detecting, warning against, and intercepting deceptive financial vectors before money changes hands.

Problem: Scammers exploit fake SEBI registration numbers, cloned broker portals, Telegram/WhatsApp pump-and-dump groups, deepfake "finfluencers," and remote-access apps. Victims typically realise it only when withdrawals are blocked.

Suggested Directions:

- **Scam & Claim Verifier** — Cross-check a social handle, message, link, or claim against known red-flag patterns and official registries in real time.
- **Tip-Group Risk Profiler** — Let a user forward a screenshot or message from a WhatsApp/Telegram "tip" channel and return an explainable risk score.
- **Mule & Phishing Radar** — Warn users before sharing credentials, granting remote-desktop access, or transferring funds to suspicious accounts.

Track B — Investor Awareness, Rights & Grievance Redressal

Focus: Making investor rights, protections, and complaint mechanisms usable by a first-time user.

Problem: Investor-protection processes such as SCORES, nominee registration, and IEPF unclaimed-dividend recovery are often too complex for someone using them for the first

time — especially senior citizens, homemakers, and legal heirs in smaller towns.

Suggested Directions:

- **Grievance Assistant** — Guide a user through framing a clear, well-documented complaint against a broker or depository participant.
- **Nominee & Family Wealth Tracker** — Help first-generation investors record nominees across Demat, bank, and mutual-fund folios to prevent unclaimed inheritance.
- **Rights & Process Navigator** — Turn a regulatory process into a short, plain-language, step-by-step journey.

Track C — Investor Education for Bharat

Focus: Replacing jargon-heavy disclosures and static PDFs with understanding-first, regional-language learning.

Problem: Financial terminology — NAV, risk, diversification, volatility, compounding, fees, nomination — is written for the financially literate, pushing first-time users toward simplified but often fake advice instead.

Suggested Directions:

- **Voice-First Explainer** — A conversational guide in regional languages that explains market and product concepts through everyday analogies, built for low-bandwidth use.
- **Consequence Simulator** — Let a user safely experience what volatility, leverage, or a market crash feels like, with zero real money at risk.
- **Plain-Language Disclosure Reader** — Digest a prospectus, SID, or CAS statement into a short summary that highlights the top risks in local language.

Track D — Financial Habits & Behavioural Resilience

Focus: Helping investors pause, reflect, and build discipline instead of acting on impulse.

Problem: Fear, greed, FOMO, herd behaviour, and revenge trading push investors — including those funding trades with emergency savings or instant loans — toward decisions they would not make with a clear head.

Suggested Directions:

- **Cooling-Off Circuit Breaker** — Detect erratic patterns (rapid loss-chasing trades, late-night impulsive activity) and introduce a deliberate pause before the next action.
- **Decision Journal** — Prompt a user to state their reasoning and time horizon before committing funds, building a habit of patient, considered decisions.
- **Goal-Anchored Tracker** — Frame investments against real life goals rather than intraday price movement.

Track E — Misinformation & Financial Content Literacy

Focus: Helping users evaluate the flood of financial content on WhatsApp, YouTube, Instagram, and Telegram.

Problem: Investors struggle to tell education apart from promotion, and confident-sounding claims apart from evidence-backed ones — often receiving no better answer than a simplistic "true/false."

Suggested Directions:

- **Claim Evidence-Checker** — Assess whether a financial claim is backed by verifiable evidence and communicate the uncertainty honestly, not as a binary verdict.
- **Promotion vs Education Classifier** — Help a user distinguish content that is educating them from content that is selling them something.
- **Source Tracer** — Trace a circulating claim back toward a reliable, official source.

Open Track — Open Innovation for Investor Resilience

Any software solution that meaningfully strengthens investor resilience and respects the Mandatory Guardrails is welcome here, even if it doesn't fit the tracks above. This track is deliberately open for ideas with strong real-world potential.

- Accessibility-first tools for elderly, low-literacy, or visually impaired investors
- Offline or low-bandwidth resilience: IVR, USSD, or missed-call based advisory verification
- Integrations with public digital infrastructure — Account Aggregator, DigiLocker, Bhashini
- Community-driven fraud-reporting or investor-protection circles for local self-help groups
- Plain-language deconstruction of corporate actions (bonus, split, buyback, delisting) or AGM resolutions

6. Target User

Solve for someone for whom existing financial systems and information are not easy to navigate — for example, a first-time investor from a Tier-2/3 city, a person more comfortable in a regional language than English, a senior citizen vulnerable to impersonation, or someone who has recently faced or nearly faced a scam. Teams should clearly identify who they are solving for and why that user's problem matters.

Persona	Profile	Core Vulnerability	What a Good Solution Does
Praveen, 22	Tier-3 graduate / gig worker, new to trading	Drawn to quick-money reels and Telegram F&O tips; trades on borrowed capital	Cooling-off nudges, fraud/finfluencer detector, risk simulator
Kavita, 39	Tier-2 homemaker managing family savings, not fluent in English	Intimidated by broking apps; susceptible to high-yield Ponzi or fake IPO schemes	Vernacular voice guidance, plain-language disclosure reader, scam radar
Babulal, 63	Retired pensioner holding old/dormant folios	Unaware of nominee process; cannot navigate online IEPF/SCORES grievance portals	Conversational grievance assistant, family nominee tracker

7. Product Expectations

- **Accessible** — simple enough for a non-expert, first-time user.
- **Inclusive** — built for regional languages, voice, and varying digital literacy.
- **Practical** — solves a problem that could realistically occur in everyday life.
- **Trustworthy** — clearly separates fact, uncertainty, and educational content.
- **Actionable** — tells the user what to safely do next, not just what is true.
- **Scalable** — has a credible path beyond a campus prototype.

8. Important Constraints

This hackathon is about investor resilience, not investment returns. A product may use financial or market data for educational, awareness, simulation, safety, or analytical purposes — but it must never turn that data into a personalised investment recommendation.

Allowed	Not Allowed
"Here is what volatility means and how it can affect an investor."	"Based on your profile, buy XYZ stock."

9. Hackathon Timeline

Dates	Phase	What Happens
27 – 29 Sep	Promotion & Onboarding	Hackathon promotion across campuses; team registration and onboarding.
30 Sep	Orientation	Teams are walked through the problem statement, tracks, and guardrails to align on scope.
1 – 4 Oct	Build Sprint	Teams work on their solution and prepare their submission.
5 Oct	Shortlisting	Top 5–7 teams are shortlisted from all submissions.
6 Oct	Final Jury Round	Shortlisted teams present before the jury.
7 Oct	Results	Top 4 teams are announced, each giving a quick presentation of their work.

(Pending council confirmation.)

10. One-Week Scope & Expected Submission

The challenge is designed to be completed within 7 days. Teams are not expected to build a production-ready platform — a convincing working prototype that proves **Problem → User → Solution → Technology → Real-world usefulness**, through at least one complete user journey, is the goal.

Each team should submit:

- **Product** — a working prototype demonstrating the core functionality.
- **Problem Definition** — who the target user is and what specific problem is being solved.
- **Solution** — how the product improves investor resilience.
- **Technology** — architecture, major technologies, AI/ML components, data sources, key design decisions.
- **Demonstration** — a 3–5 minute demo showing a realistic user scenario.
- **Impact** — who benefits, what harmful behaviour it reduces, why it matters for Tier-2/3 India, and how it could scale.

11. Evaluation Criteria

Criterion	Weight	What We Look For
Resilience & Safety Impact	30%	Does the product measurably help a user avoid fraud, avoid loss, or build safer financial behaviour?
Tier-2/3 Usability (Bharat-First)	25%	Regional-language support, low-bandwidth and low-end-device readiness, voice/visual UX, minimal cognitive load.
Guardrail Compliance & Trust	15%	Strictly non-commercial; no stock tips or conflicts of interest; transparent about uncertainty; user privacy preserved.
Technical Execution	15%	Technology (AI/ML, NLP, voice, on-device processing) used effectively rather than added for novelty.
Feasibility & Scalability	15%	Could this realistically extend beyond the hackathon into a tool used by real investors?

12. The Bigger Question

We are not asking *"how can we help people invest better?"*

We are asking: how can technology help Indians become harder to fool, easier to educate, better prepared, and more resilient when making financial decisions?

Build something that could sit in the hands of a first-time investor in Varanasi, Ranchi, Jaipur, Guwahati, Indore, Patna, Coimbatore, or any other emerging Indian city — and genuinely make that person's financial journey safer and more informed.